Data Cleaning, Transformation and Modeling Data Assignment

Transform data link= https://drive.google.com/drive/folders/1xog4J-Qp9OIkXlzCQGuSWC-w15Q12Pew?usp=sharing

Question 1: What is Data Loading in Power BI and why is it considered the first step of analysis?

Answer

Data loading is the process of importing the data into power BI and create the reports and analysis.

Power BI allows:

- Direct data loading.
- Transform the before load.

Step to load data into power BI

- Open power BI desktop
- Click “GET DATA”.
- Select the data sources.
- Preview & load data.

Question 2: Explain the difference between “Load” and “Transform Data” in Power BI.

Answer

Load

Import data directly into power BI

No modifications are made.

Faster process.

Example: import csv as it is.

Transform data

Open power query editor for data cleaning.

Data can filter, cleaned, transformed

Used before loading data into the model.

Example: Remove null, duplicates, change data type.

Question 3: What is a Fact Table and a Dimension Table? Give examples from the dataset.

Answer

Fact table (center).

Stores transactional data.

Contains numeric measures and foreign keys.

- Confirmed_Cases
- Recovered_Cases
- Deaths
- Active_Cases
- Tests_Conducted
- Hospitalized

Dimension table (normalised)

Split into multiple related tables

Example: product – product category – product sub category

- Country
- State
- Date
- Vaccination_Status

Question 4: Why is Star Schema preferred over Snowflake Schema in Power BI? Practical.

Answer

Star Schema

Simple easy to understand.

Faster query performance.

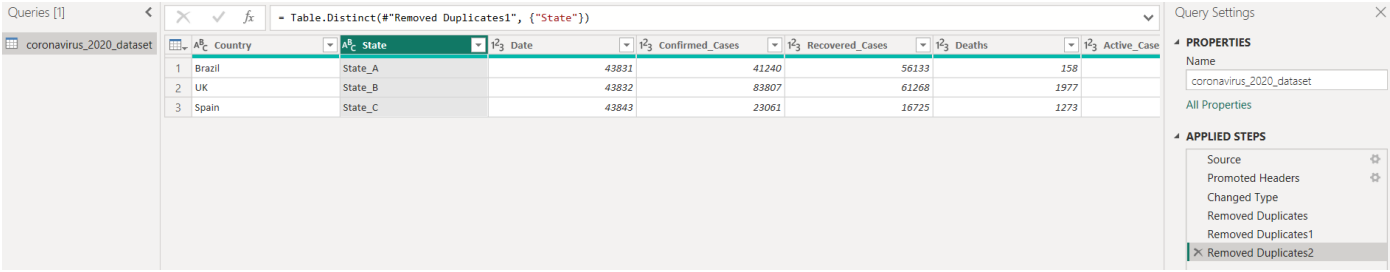
Fewer relationship issues.

Best for slicing and filtering data.

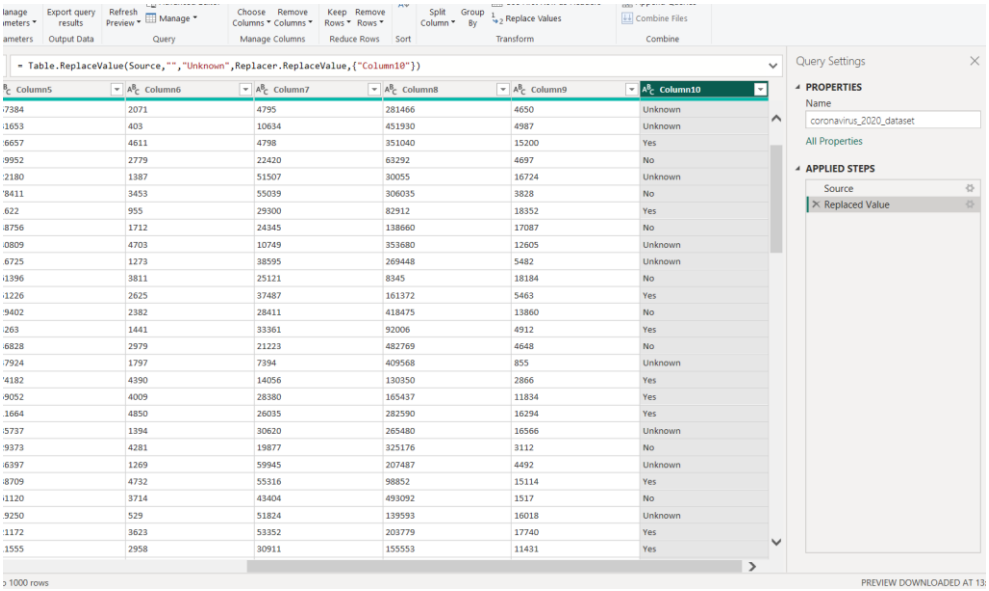
Improve report performance.

Power BI generally recommends a Star Schema for data modeling.

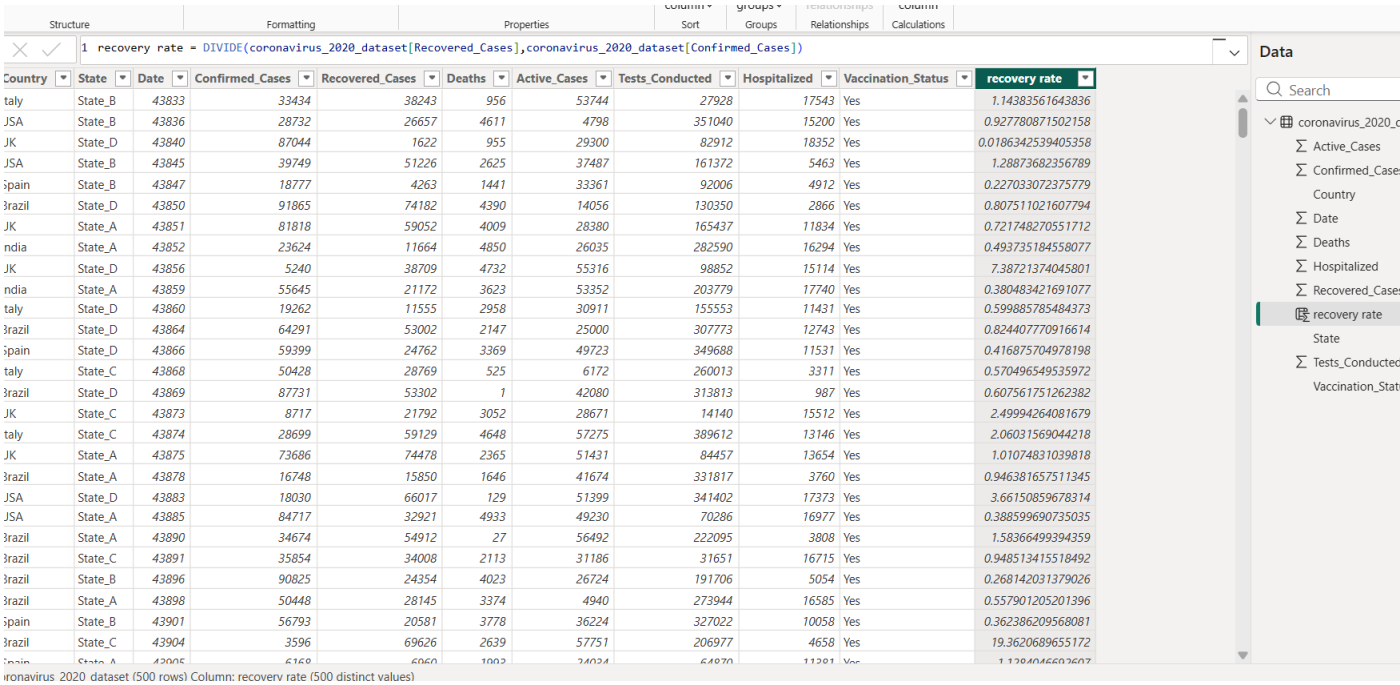
Question 5: Identify and remove duplicate records based on Date, Country, and State.



Question 6: Identify and replace null values in Vaccination_Status.



Question 7: Create a new column to calculate Recovery Rate.



Question 8: Create a summarized table showing total confirmed cases by Country.

